12th HACKERS KA JUGAAD

Nuclei

Composition of nucleus

The atom consists of central nucleus containing entire positive charge and almost entire mass. According to accepted Model the nucleus is composed of protons and neutrons. The proton was discovered by Rutherford by bombardment of alpha particles on nitrogen in accordance with the following equation.

$$^{14}_{7}N + {}^{4}_{2}He \longrightarrow {}^{17}_{8}O + {}^{1}_{1}H$$

Symbolically a nuclide is written as ${}^{A}_{Z}X$ or ${}_{Z}X^{A}$ where A is the mass number of Z is the atomic number

The neutron was discovered by J Chadwick by bombardment of alpha particles on beryllium in accordance with

$$^{9}_{4}Be + {}^{4}_{2}He \longrightarrow {}^{13}_{6}C + {}^{1}_{0}n$$

- A neutron is neutral particle and its mass number is 1
- The number of protons in the nucleus is called atomic number. While the number of nucleons is called the mass number.
- In general mass number is greater than atomic number except for hydrogen nucleus, where A equals Z
- Since neutron is neutral, it is used for artificial disintegration

Size of nucleus

According to experimental observations, the radius of the nucleus of an atom of mass number A is

$$R \propto A^{1/3} \qquad R = R_0 A^{1/3}$$

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi (R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0^3 A.$$

$$\rho = \frac{m}{V} = \frac{mA}{4/3\pi R_0^3 A}.$$

$$\rho = \frac{3m}{4\pi R_0^3} = \text{constant}.$$

Atomic Masses

The masses of atoms, nuclei, etc., are expressed in terms of atomic mass unit represented by amu or 'u'. For this mass of C-12 is taken as standard.

$$1\,u = \frac{\text{mass of carbon} - 12 \text{ atom}}{12}$$

$$1u = 1.1660565 \times 10^{-27}\ kg$$

- mass of proton (m_p) = 1.007276 u
- mass of neutron (m_n) = 1.008665 u
- mass of electron (m_e) = 0.000549 u

Isotopes, Isobars and Isotones

- The nuclides having the same atomic number, but different mass number are called isotopes
- The nuclides having the same mass number, but different atomic number or called isobars
- The nuclides, having the same number of neutrons are called isotones

Mass energy equivalence Reaction

According to Einstein, the mass and energy are equivalent i.e., mass can b converted into energy and vice-versa. The mass energy equivalence relation is

$$E = mc^2$$

Accordingly, 1 kg mass is equivalent to energy = $1 \times (3 \times 10^8)^2 = 9 \times 10^{16}$ joules and

$$1 \text{ amu} = \frac{1}{6.02 \times 10^{26}} \text{ kg}$$

mass is equivalent to energy 931 MeV.

1amu = 931 MeV

Mass defect

It is observed that the mass of a nucleus is always less than the mass of constituent nucleons (i.e., protons + neutrons). This difference of mass is called the mass defect.

Let (Z, A) be the mass of nucleus, m_p = the mass of proton and m_n = mass of neutron, then the mass defect

$$\Delta m = \text{Mass of nucleons} - \text{Mass of nucleus}$$

$$\Delta m = Zm_p + (A - Z)m_n - M_{\text{nucleus}}$$

Binding energy per nucleon

This mass defect is in the form of binding energy of nucleus, which is responsible for binding the nucleons into a small nucleus.

Binding energy of nucleus = $(\Delta m)c^2$

Binding energy per nucleon = $\frac{(\Delta m)c^2}{A}$.

Nature of nuclear forces

- The protons and neutrons inside the nucleus are held together by strong attractive forces. These attractive forces cannot be gravitational since forces on repulsion between protons >> attractive. Gravitational force between protons. these forces are short-range. Attractive forces called nuclear forces .

- The nuclear forces are strongest in nature, short range and charge independent. Therefore the force between proton proton is the same as the force between neutron neutron or proton neutron.

Nuclear Reaction

When a beam of monoenergetic particles (e.g., α-rays, neutrons etc.) collides with a stable nucleus, the original nucleus is converted into a nucleus of new element. This process is called a nuclear reaction. A typical nuclear reaction is

$$a + X \rightarrow Y + b$$

where a is incident energetic particle, X is target nucleus, Y is residual nucleus and b is outgoing particle.

- In a nuclear reaction mass number, electric charge, linear momentum, angular momentum and total energy are always conserved. The energy of reaction is

$$Q = (M_a + M_X)c^2 - (M_b + M_Y)c^2$$

$$Q = (M_{reactant} - M_{product})c^2$$

Nuclear Fission

The splitting of heavy nucleus into two or more fragments of comparable masses, with an enormous release of energy is called nuclear fission. For example, when slow neutrons are bombarded on $_{92}U^{235}$, the fission takes place according to reaction

$$^{235}_{92}U + \underset{\text{(slow neutron)}}{^{1}_{0}n} \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3(^{1}_{0}n) + 200\ \text{MeV}$$

In nuclear fission the sum of masses before reaction is greater than the sum of masses after reaction, the difference in mass being released in the form of fission energy.

Remarks:

1. It may be pointed out that it is not necessary that in each fission of uranium, the two fragments Ba^{141} and Kr^{92} are formed but they may be any stable isotopes of middle weight atoms.

2. The fission of U^{238} takes place by fast neutrons.

Nuclear Fusion

The phenomenon of combination of two or more light nuclei to form a heavy nucleus with release of enormous amounts of energy is called nuclear fusion. The sum of masses before fusion is greater than the sum of masses After fusion, the difference in mass appearing as fusion energy

For example diffusion of two deuterium nuclei into helium is expressed as $${}_1^2H + {}_1^2H \longrightarrow {}_2^4He + 21.6\,MeV,$$

Thus fusion process occurs at an extremely high temperature and high pressure as is sun where temperature is 10^7 K

Remarks –

- For the fusion to take place, the component nuclei must be brought within a distance of. 10^{-14} m for this, they must be imparted high energies to overcome the repulsive force between nuclei. This is possible when temperature is enormously high
- The principle of hydrogen bomb is also based on nuclear fusion
- The source of energy of sun and other star is nuclear fusion. There are two possible cycles

Proton- proton cycle

$$_1^1H + {}_1^1H \longrightarrow {}_1^2H + {}_1^0\beta + \nu. \text{ (Neutrino)} + \text{Energy}$$

$$_1^2H + {}_1^1H \longrightarrow {}_2^3He + {}_0^1n + \text{Energy}$$

$$_2^3He + {}_2^3He \longrightarrow {}_2^4He + {}_1^1H + {}_1^1H + \text{Energy}$$

Net Result is $_1^1H + {}_1^1H + {}_1^1H + {}_1^1H \longrightarrow {}_2^4He + 2\,{}_1^0\beta + 2\upsilon + \text{Energy (26.7 MeV)}$

Carbon– Nitrogen Cycle

$$_1^1H + {}_6^{12}C \longrightarrow {}_7^{13}N + \text{Energy}$$

$$_7^{13}N \longrightarrow {}_6^{13}C + {}_1^0\beta + \nu \text{ (neutrino)}$$

$$_6^{13}C + {}_1^1H \longrightarrow {}_7^{14}N + \text{(Energy)}$$

$$_7^{14}N + {}_1^1H \longrightarrow {}_8^{15}O + \text{Energy}$$

$$_8^{15}O \longrightarrow {}_7^{15}N + {}_1\beta^0 + \nu \text{ (neutrino)}$$

$$_7^{15}N + {}_1^1H \longrightarrow {}_6^{12}C + {}_2^4He + \text{Energy}$$

Net result is $_1^1H + {}_1^1H + {}_1^1H + {}_1^1H \longrightarrow {}_2^4He + 2\,{}_1^0\beta + 2\nu + \text{Energy (26.7 MeV)}$

- The proton-proton cycle occurs at a relatively lower temperature as compared to carbon-nitrogen cycle which has a greater efficiency at higher temperature.
- At the sun whose interior temperature is about 2×10^6 K, the proton-proton cycle has more chances for occurrence.

THANK YOU !!